



More Complex Factoring

When you factor a trinomial of the form $ax^2 + bx + c$ when $a > 1$, you will have to use trial and error.

Example

Factor: $2x^2 + 9x + 4$

Step 1

Find the factors of $2x^2$.

$2x \cdot x = 2x^2$

Step 2

List the factors of 4.

4 and 1

2 and 2

Step 3

Use trial and error to find the correct factor pair

$(2x + 4)(x + 1) = 2x^2 + 6x + 4$

$(2x + 1)(x + 4) = 2x^2 + 9x + 4$

$(2x + 2)(x + 2) = 2x^2 + 6x + 4$

Step 4

Write the factors.

$(2x + 1)(x + 4)$

The factorization of $2x^2 + 9x + 4$ is $(2x + 1)(x + 4)$.

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